DESIGN AND IMPLEMENTATION OF EARLY DETECTION AND MANAGEMENT OF MAIZE CROP DISEASES USING MACHINE LEARNING

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A PROJECT SUBMITTED TO THE DEPARTMENT OF SOFTWARE ENGINEERING, FACULTY OF COMPUTING, NILE UNIVERSITY OF NIGERIA, IN PARTIAL FULFILLMENT OF THE REQUIREMENTS FOR THE AWARD OF DEGREE OF BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY,

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DEDICATION

This project is dedicated to Almighty God for His guidance, strength, and grace throughout the course of this degree program.

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ABSTRACT

Maize production remains a cornerstone of global food security; however, diseases such as Northern Leaf Blight and Common Rust can cause yield losses of up to 40%. Although Machine Learning (ML) techniques have shown promising results in controlled environments, many existing solutions remain inaccessible to smallholder farmers due to usability and accessibility limitations. This research presents the design and implementation of an integrated web-based system for the early detection and management of maize diseases. Using a Convolutional Neural Network (CNN), specifically the MobileNetV2 architecture, alongside a mixed-method research approach and the Waterfall development methodology, the study developed a solution capable of providing real-time disease diagnosis and management recommendations. The system emphasizes accessibility by supporting four languages: English, Hausa, Yoruba, and Igbo, enabling wider adoption among local users. Results indicate that the model effectively classifies major maize leaf diseases from uploaded images while providing actionable management recommendations to reduce excessive chemical dependence. By integrating automated image processing with a user-centered interface, this study contributes to a scalable precision agriculture framework that empowers farmers to make data-driven decisions, improve crop protection, and promote sustainable farming practices.

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CHAPTER ONE

INTRODUCTION

1.1 Background of the Study

Zea mays L. or maize is one of the most popular crops of cereals grown and consumed worldwide, playing a strategic role in the food security, industrial raw materials, and animal feed (Adesoba, 2025). Maize plays a significant role in food supply in most countries in Africa such as Nigeria and farmers income. Although the role is significant, maize production faces an increasing threat of a number of foliar diseases, including common rust, gray leaf spot, northern leaf blight, and maize streak virus that can heavily affect the yield without people realizing them and acting timely (Dube and Mambudzi, 2025; Khan *et al*., 2023).

Conventionally, maize diseases have been detected using manual scouting, as well as through the visual examination of symptomatic leaves by experts. It is usually slow and subjective and requires agricultural specialists to be available, which is not feasible both with smallholder farmers and large agricultural enterprises (Idress *et al*., 2024). Additionally, the late identification tends to cause the transmission of infections in the fields causing tremendous losses in yield and quality.

Over the past years, machine learning (ML) and deep learning (DL) methods have become one of the potent tools to be used to automate the process of identifying and classifying plant diseases. The computational methods are based on the principles of pattern recognition and image analysis to detect the signs of diseases based on a digital photograph of a crop leaf, making the process of timely and pertinent data collection possible and out of reach of traditional, manual methods (Zainuddin *et al*., 2023).

As an example, one of the most popular types of deep learning networks, convolutional neural networks (CNNs) have been highly utilized to classify maize leaf images with high accuracy, frequently over 90% in controlled experimental conditions (Nature Scientific Reports, 2022).

The idea to incorporate ML models into real-world applications like mobile apps or web platforms opens the potential of real-time, in-the-field disease diagnostics, which can be used even by farmers with little technical skills. CNNs have been successfully deployed into mobile systems to guide farmers on disease detection and management measures, which has been shown to help farmers in early intervention and crop protection (Dube and Mambudzi, 2025). On the same note, more sophisticated architectures that integrate attention and multi-scale feature extraction have continued to increase the accuracy of maize disease models, reflecting the high rate at which ML solutions in agriculture have evolved (Zhu *et al*., 2024).

In the case of the proposed project therefore, the concern shifts to the utilization of the gap between model development and implementation into mobile-software. An appropriately designed web-based app capable of inference on the device, providing a farmer with an intuitive user interface and operational behavior in a low-network or hardware performance context can go a long way in increasing the usability and effectiveness of disease-detection-based technology in actual farming conditions.

Through the design and implementation of such web-based system, the project aims to go beyond the algorithmic proof-of-concepts and provide the full software solution that is equivalent to the real-world working environment of maize farmers. This paper is part of the current research in precision agriculture where new machine learning system is developed and tested to identify and control maize crop diseases at an early stage with a focus on precision, usefulness and practical use.

1.2 Statement of Problem

Zea mays L. maize is a major staple crop in the whole world but the diseases which severely affect it include the leaf blight, rust, and gray leaf spot that can lead to a reduction in yield of up to 40% without early detection (Ngugi, Akinyelu, and Ezugwu, 2024; Idress, Arshad, and Khan, 2024). Even though recent research works have used artificial intelligence (AI) and machine learning (ML) to detect crop diseases with impressive accuracy on models like CNN and MobileNetV2 (Kallur, Yaligar, and Thotad, 2024; Olayiwola, 2023), many of these solutions are limited to the research environment and are based on offline processing.

Current systems are hardly ever transformed into fully integrated, web-based applications that would provide the opportunity to detect and manage maize diseases in real-time and on-device (Rao Bangole, 2024; Khamkar, Gawali, and Mane, 2025). Also, minimal concerns have been placed on the concepts of software engineering that are required to make such systems accessible to farmers in the real-life context, including usability, optimization, and experience (Alsuwaidi, Talib, and Nasir, 2025).

Existing integrated web-based systems are not integrated to utilize machine learning and image processing to detect maize crop disease at an early stage and manage them in real-time, especially those that would be accessible and allow their use by small holder farmers.

1.3 Aim and Objective of the Research

1.3.1 Aim of the Study

To develop an early detection and management system for maize crop diseases using machine learning.

1.3.2 Objectives of the Study

The particular objectives are to:

i. To design a system for identifying maize leaf diseases using CNN.

ii. To implement a web-based application that can be easily accessible for diagnosis.

iii. To test the functionalities of the system using user acceptance testing.

1.4 Significance of the Study

The importance of this study is that it helps to enhance the development of precision agriculture and smart farming by implementing machine learning methods in managing crop diseases. The system can help farmers make sound and data-driven decisions that can dramatically reduce crop losses and increase overall productivity by allowing farmers to identify the diseases in their maize crop and take suitable actions at the earliest possible time. The timely intervention by early diagnosis helps in preventing the spread of diseases in large areas of the farms hence reducing the economic losses that occur in case of late detection.

The system eliminates the excessive reliance on agricultural specialists and extension officers, which are usually scarce and inaccessible particularly in the rural agricultural communities. The image-based analysis of disease detection can be automated, and with this automation, farmers will be able to diagnose crop health conditions faster and more accurately on their own. This does not only make them more efficient but also equips the smallholder farmers with the modern technological tools, which they could not access before.

Moreover, the study encourages sustainable farming practices through advocacy of effective disease management practices that limit the high application of chemical pesticides. Early and precise diagnosis of diseases will mean that suitable control strategies are implemented only when required hence the environment is sustainable and food security enhanced. Academically, the study also contributes to the existing literature on the implementation of machine learning in agriculture and offers a feasible framework of future studies and systems advancement.

1.5 Scope of the Study

The proposed project is aimed at designing and developing a web-based and Android-enabled system to detect and manage maize crop diseases at an early stage with the help of machine learning tools. The paper is confined to image-based disease detection, in which operators take pictures of maize leaves with the help of an Android-based mobile phone to be analyzed in real-time.

The system is particularly specific to the attack of the maize diseases that are major and prevalent, such as the Northern Leaf Blight, Gray Leaf Spot, and Common Rust. A Convolutional Neural Network (CNN) is used to classify the disease, as it is the most effective in terms of image recognition. The trained model is optimized to work on mobile devices and is trained with labeled datasets of maize leaf images.

The system has offline capability to increase the usability in rural and lowconnectivity regions where it enables one to detect the disease without regular internet connection. The other maize health issues that include the lack of nutrients, pests, and soil related are however not discussed in the study. The scope is also narrowed to detecting an illness and elementary management guidelines instead of general administration or production forecasting.

1.6 Limitations of the Study

Despite the strengths of this study, several limitations were identified. The performance of the system is highly dependent on the quality of images captured by users. Factors such as poor lighting conditions, blurry images, and improper leaf positioning can negatively affect the accuracy of the model.

Additionally, the system is limited to identifying maize diseases that are represented within the training dataset and may not accurately classify rare or previously unseen diseases. The CNN model was trained using existing datasets, which may not fully represent all environmental conditions encountered in real farming environments. Variations in climate, maize species, and disease presentation across different regions may therefore influence detection accuracy.

Furthermore, although the system was designed to operate effectively under low-network conditions, periodic model updates and retraining may still require internet access and technical support to maintain optimal performance.

1.7 Organisation of the Study

This study paper will be divided into five chapters. Chapter One presents the study by giving the background, problem statement, objectives, significance, scope, limitations and definitions of the key terms. The second chapter provides a review of the literature concerning the topic of maize crop diseases, machine learning approaches, and associated disease detection systems. Chapter Three outlines the approach taken in the system design and implementation such as data collection, development of the model and system architecture.

The fourth chapter includes the system implementation, experimental outcomes, performance analysis and discussion of findings. Lastly, Chapter Five concludes the study with the summary of the findings, and contributions and recommendation on future research.

1.8 Definition of Key Terms

- Maize (Zea mays): A cereal crop widely cultivated for food, animal feed, and industrial purposes.
- Machine Learning: A branch of artificial intelligence that enables systems to learn patterns from data and make predictions without explicit programming.
- Convolutional Neural Network (CNN): A deep learning algorithm particularly effective for image processing and classification tasks.
- Early Disease Detection: The identification of crop diseases at an initial stage before severe symptoms appear or spread.
- Precision Agriculture: A farming approach that uses technology and data analysis to optimize crop production and resource use.
- Smart Farming: The integration of digital technologies such as sensors, artificial intelligence, and automation into agricultural practices.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

The chapter is a literature review of the topic of Design and Implementation of Early Detection and Management of Maize Crop Diseases Using Machine Learning. It introduces the main concepts, technical designs, machine learning models, image treatment methods and real-time deployment methods that underlie the contemporary smart farming systems. This chapter is a good theoretical and technical foundation of the proposed system as it organizes previous research into major thematic settings and the essential gaps in the research that the present project will fill.

2.2 Maize Production and Its Significance Overview

Maize is a pillar in world and domestic agriculture that serves as the force behind food security, economic development, and rural livelihoods and poses a challenge in sustainability. It is the largest cereal crop in terms of the volume of production and the number of hectares under cultivation with an annual production of over 1.1 billion tons (Erenstein *et al*., 2022; Păcurar *et al*., 2025; Shiferaw *et al*., 2020). More than 4.5 billion people use maize to supply over 30% of their food calories, and thus it is a vital part of the global diet and food security (Shah *et al*., 2025; Shiferaw *et al*., 2021). It is utilized in various ways: it is eaten by people in subSaharan Africa, Latin America, and some areas of Asia as a primary food product, as a source of biofuels, and raw material industries in developed economies (Erenstein *et al*., 2022; Ranum *et al*., 2022; Tanumihardjo *et al*., 2020). The growing need associated with population increase, urbanization, and dietary changes accentuate the significance of maize in the world even more (Erenstein *et al*., 2021; Shah *et al*., 2025).

Maize supports rural livelihoods, food security, and jobs, especially to smallholder farmers in Africa, Asia, and Latin America at the local level (Mdoda *et al*., 2025; Martey *et al*., 2020; Busungu, 2025). It is the primary source of calories in various areas, as it offers food and a source of income to millions of families (Shiferaw *et al*., 2021; Shah *et al*., 2025). The use of better maize types is reported to augment the yield and technical performance that also lead to increment in household incomes and environmental resilience (Manda *et al*., 2020; Mdoda *et al*., 2025; Asante *et al*., 2024). Nevertheless, productivity gains are limited by other issues like the lack of access to quality seeds, lack of an appropriate infrastructure, and ineffective extension services (Martey *et al*., 2020). Maize production has also substituted the conventional crops in other areas transforming diets and agriculture (Busungu, 2025).

Besides the economic and nutritional significance, maize helps to maintain sustainability and climate resilience. Better species and adaptive management increase productivity in the shifting climatic conditions (Ahmad *et al*., 2020; Cammarano *et al*., 2020; Nassar *et al*., 2025). However, environmental concerns continue to be degradation of land, herbicides leaching, and soil fertility (Lairez *et al*., 2023; Păcurar *et al*., 2025). The need is to balance the productivity with environmental stewardship through sustainable production practices, policy support and further research (Erenstein *et al*., 2022; Tanumihardjo *et al*., 2020).

2.3 Maize Crop Diseases and their effects

Maize is very susceptible to many diseases and pests which lower the yield, quality of the grain and income of the farmer globally. Some of the most devastating ones are Maize Lethal Necrosis (MLN), Gray Leaf Spot (GLS), Northern Leaf Blight (NLB), Southern Corn Rust (SCR), Tar Spot Complex (TSC) and Maize Ear Rot. These organisms are fungi, viruses, and insect pests that cause diseases that also have regional and environmental differences (Njeru *et al*., 2023; Redinbaugh and Stewart, 2020; Mahuku *et al*., 2021).

As an example, in extreme cases, the total yield of crops can be completely lost by MLN, whereas in Africa and Asia, the annual losses in production are up to one million tonnes due to fall armyworm attacks (Flora *et al*., 2023; De Groote *et al*., 2020). In temperate and tropical areas, fungus diseases including GLS and NLB are challenges that continue to reduce yields by an average of 10%-33%(Nsibo *et al*., 2024; Terefe *et al*., 2023).

All these diseases lead to the loss of maize in the world, ranging between 19.5%–41.4% per year, equivalent to serious economic and food security problems (Savary *et al*., 2020; Wang *et al*., 2024). The annual losses in North America alone are about US138/hectare because of diseases of maize (Mueller *et al*., 2020). In addition to the reduction in yields, the diseases such as the Maize Ear Rot inject mycotoxins that are detrimental to the food chain which is a health and trade hazard (Kusa *et al*., 2025). The synergies of these biotic stresses is most catastrophic to the smallholder farmers who are heavy users of maize as a staple crop, which poses an additional threat to the food security and livelihood of the region (Mahuku *et al*., 2020; Njeru *et al*., 2023).

The current disease diagnosis system in maize crops has a number of constraints that make it hard to detect and manage the disease in time. The most common approach is manual visual inspection, which is relatively time-consuming and labor-intensive and, therefore, cannot be used to monitor diseases on a large scale or in a short time frame (Jafar *et al*., 2024; Zhang *et al*., 2020). The method relies greatly on human judgment which brings subjectivity and lack of consistency. The differences in the level of skills and environmental conditions commonly cause negative diagnosis or poor attention to early symptoms (Tugrul *et al*., 2022; Jafar *et al*., 2024).

The lack of scalability is another limitation that limits the performance of conventional approaches. In large or distant farms of maize, the data cannot be sufficiently inspected manually, which slows the diagnosis of the disease and makes it possible to be infected by many people (Zhang *et al*., 2020).

Traditional diagnosis is also inclined towards the detection of diseases when visible symptoms occur, usually at mid and late stages of infection, when the loss of yields is already high (Zhang *et al*., 2020). In addition, it is costly and demanding in terms of resources, such as time, labor, and expert knowledge, which is why it is inaccessible to many smallholder farmers (Jafar *et al*., 2024).

2.4 Artificial Intelligence and Agriculture

The concept of Artificial Intelligence (AI) has become the revolutionary power in the agricultural industry, which essentially improves the precision, automation, and data-driven decision-making (Subeesh and Mehta, 2021; Benos *et al*., 2021). Combining machine learning (ML) and deep learning (DL) models, farmers will be able to streamline production processes and cut costs and ensure sustainable use of resources in different areas of monitoring and management (Soussi *et al*., 2024).

In particular, AI systems can be used in herbal health management, which greatly increases the efficiency of this field, allowing efficient, quick, and large-scale detection and prediction of infection spread (Shoaib *et al*., 2023; Jafar *et al*., 2024; Hasan *et al*., 2020). ML and DL models are used to process visual plant images and other sensor data to diagnose diseases more precisely and faster than traditional methods. This would also be essential because in this way, systems can detect infections at their initial stages when they may remain unnoticeable, thus significantly reducing outbreaks and preventing losses through the harvest of crops in time (Dhaka *et al*., 2021; Abade *et al*., 2020). The transition to AI is a decisive step in attaining more sustainable and efficient agricultural results.

2.5 Disease Identification by Image Processing Techniques

2.5.1 ML in Computerized Crop Disease detection

Machine learning (ML) is regarded as a fundamental technology in contemporary agriculture, and it is being used to provide automated, expedited, and precise techniques of detecting crop diseases. ML models can improve early diagnosis and aid in precision agriculture through the analysis of images and sensor data (Jafar *et al*., 2024; Shoaib *et al*., 2023; Benos *et al*., 2021).

2.5.2 Classification of diseases based on images

Convolutional Neural Networks (CNNs) and other types of deep learning algorithms are popular in the classification and detection of crop diseases based on the image of a leaf, stem, or fruit. These models distinguish the healthy and infected plants, the types of diseases and the degree of severity. They are usually faster, more accurate, and scalable than manual inspection (Jafar *et al*., 2024; Shoaib *et al*., 2023; Ngugi *et al*., 2020; Liu and Wang, 2021; Dhaka *et al*., 2021; Tugrul *et al*., 2022; Abade *et al*., 2020; Benos *et al*., 2021).

2.5.3 Data Preprocessing and Data Collection

Quality datasets are required to have an effective ML-based disease detection. The collection of data is performed by drones, IoT, smartphones, and satellites with the help of RGB and hyperspectral images. Preprocessing: Preprocessing methods (segmentation, image enhancement, and feature extraction) are useful to isolate diseased regions to enhance the accuracy and reliability of the model (Jafar *et al*., 2024; Ouhami *et al*., 2021; Shoaib *et al*., 2023; Liu and Wang, 2021; Benos *et al*., 2021).

2.5.4 Early Detection and Automated Monitoring

ML models will allow monitoring of crops in very high numbers and in real-time. They detect infections at early or even pre-symptomatic stages, which enables farmers to intervene before extensive damages take place. This lowers the losses, restricts the pesticides and promotes sustainable agriculture (Domingues *et al*., 2022; Jafar *et al*., 2024; Shoaib *et al*., 2023; Ngugi *et al*., 2020; Liu and Wang, 2021; Benos *et al*., 2021).

2.5.5 Connection to Decision Support Systems

Farm management platforms can be integrated with machine learning in order to come up with alerts and treatment suggestions in a timely manner. Such integration enhances the use of data to make decisions, fostering the use of resources efficiently and accuracy of the control of the diseases (Domingues *et al*., 2022; Ouhami *et al*., 2021; Shoaib *et al*., 2023; Benos *et al*., 2021).

2.6 Related Works

The combination of machine learning (ML), deep learning (DL), and image processing has been important in detecting and controlling the disease of maize crops earlier and before it becomes untreatable. The technologies have increased the level of accuracy, scaling, and real-time responsiveness in the identification and monitoring of crop diseases. Different researches helped to design and implement AI-based systems that are used to automate the process of diagnosing diseases, help with precision agriculture, and enhance the decision-making process.

Abade *et al*. (2020) and Hasan *et al*. (2020) have investigated the application of deep learning, especially Convolutional Neural Networks (CNNs) in the process of detecting maize leaf disease. Their models trained on UAV and ground-based images datasets had a precision in classification of over 95%. These CNN models performed well in the extraction of spatial and contextual features, which led to the strong detection of disease. Nevertheless, the two studies also emphasized the main shortcomings as the necessity of large annotated data and its decline of performance in different environmental conditions.

More development was made possible by way of multi-source data fusion. Ouhami *et al*. (2021) and Jafar *et al*. (2024) came up with systems that combined spectral and environmental data along with visual data by the use of IoT sensors, UAVs, and satellite imagery. These models improved the level of prediction of diseases at the early stage and allowed real-time monitoring. Although these existed, they were constrained by the computational issues, network complexity as well as the need to integrate the data in various sources.

Recent studies have been done on the creation of end-to-end AI models used to detect and manage diseases automatically. Shoaib *et al*. (2023), Liu and Wang (2021), and Jafar *et al*. (2024) suggested completely automated pipelines covering image acquisition, preprocessing, segmentation, feature extraction and classification. These systems reduced the level of manual work which ensured real time analysis and this made them effective in the large-scale monitoring of agriculture. Nevertheless, they were constrained with regard to the model generalization and flexibility in other regions, varieties of maize, and stages of the disease.

In addition to this, Ahmad *et al*. (2022) stated the necessity of standardized datasets, field validation, and decision-support system integration. Another recommendation in the study was that greater cooperation between AI developers and agricultural experts should be considered to make the results of research practical in terms of their application in the field.

A number of more recent publications have specifically touched upon the design and implementation of AI-based systems to detect maize disease. Kallur and Prakash (2023) trained a model of maize disease detection based on a web-based architecture with the YOLOv8n architecture to attain a 99.04% accuracy in identifying and localizing the disease in real-time. On the same note, Khan *et al*. (2024) adopted a CNN model that is augmented with transfer learning and data augmentation to attain 98.6% accuracy with computational efficiency to be deployed on the web. Haque *et al*. (2022) used the Inception-v3 and ResNet CNNs on images taken in the field and obtained an accuracy of 95.99% and adaptation to different lighting and backgrounds.

Adebayo and Li (2023) modeled a 15-layer CNN model that is optimized to achieve a high computational rate and fewer parameters without affecting its accuracy. In a related development, Zhao and Cheng (2024) used Vision Transformer (ViT) architectures and in this case, the MaxViT architecture, where the detection accuracy was over 99% on large and diverse datasets. Chukwuma *et al*. (2022) also combined weakly supervised learning with higher interpretability and localization of disease spots using web applications.

Previous methods used the traditional ML models like Support Vector Machine (SVM), the Random Forest (RF), and Naive Bayes (NB) models with handcrafted color, texture, and shape features. Accuracy using these classical methods was found to be 90-92.7% by Osei and Boateng (2018). Nevertheless, recent comparative analyses (Haque *et al*., 2022; Khan *et al*., 2024) all indicate that the DL-based methods are more precise and significant in terms of scalability than the traditional ones.

Besides detection, there has been an attempt to increase management capacities. Ahmad *et al*. (2022) proposed a system of IoT-enabled deep learning that can simplify monitoring and triggering alerts and Prakash and Kirubakaran (2023) suggested systems that can offer implementable advisory information on disease control. Indeed, these inventions go beyond single point detection to combination of disease management.

On the whole, the current research proves that AI and ML have a high potential in the detection and control of maize disease. The major successes consist of a high diagnostic classification, real time implementation and better scalability. However, several issues still exist, including the lack of diversity in the datasets, the high computing cost, the absence of the validation of the models in the field, and the inability to incorporate AI models into platforms accessible to farmers. It will be critical to curb these challenges in pursuit of efficient, convenient, and sustainable AI-based agriculture.

2.7 Summary of Literature Review

Title / Focus Area	**Author(s) and Year**	**Findings / Achievements**	**Research Gaps / Limitations**
Maize Leaf Disease Detection and Classification Using Machine Learning Algorithms	Panigrahi, K. P., Das, H., Sahoo, A. K., and Moharana, S. C. (2020)	Random Forest achieved the highest accuracy (79.23%), helping farmers detect maize diseases early.	Dataset size, disease diversity, and computational limits not addressed; moderate accuracy indicates room for improvement.

Maize Leaf Disease Detection Using Convolutional Neural Network (CNN)	Sharma *et al.* (2021)	CNN model accurately identified maize leaf diseases under controlled conditions.	Performance dropped with real-field images due to lighting and background variations.
Deep Learning Approach for Automated Detection of Plant Diseases	Ahmed and Singh (2020)	Demonstrated efficiency of AI in plant disease recognition using deep learning.	Focused on general plant diseases, not maize-specific; limited dataset diversity.

Machine LearningBased Model for Early Detection of Maize Leaf Diseases	Bello *et al.* (2022)	Classified maize diseases at early stages, enabling early intervention.	Required large, high-quality datasets; limited real-world validation.

AI-Powered Crop Health Monitoring System Using Image Processing	Kumar and Adewale (2023)	Combined AI and IoT for real-time crop health assessment.	High computational cost; dependent on connectivity and hardware.
Comparative Study of Machine Learning Algorithms for Crop Disease Prediction	Li and Osei (2019)	Compared SVM, RF, and CNN; CNN achieved best prediction accuracy.	Did not include real-time management or mobile integration.

Limited Real-World Validation and Scalability of AI Models	Olayiwola (2023); Mall *et al*. (2023); Kallur *et al*. (2024); Rao Bangole (2024); Alsuwaidi *et al*. (2025)	AI models achieved high accuracy in controlled settings but reduced reliability in field conditions.	Few models validated under diverse conditions; need scalability and adaptability improvements.
Lack of Explainability and User Interaction in AI Models	Banadda *et al*. (2023); Rao Bangole (2024)	Many AI systems operate as black boxes, limiting user trust.	Need for explainable AI (XAI) and interactive feedback

Limited Coverage of Diseases and Crop Types	Olayiwola (2023); Khamkar *et al*. (2025); Kallur *et al*. (2024)	Focused mainly on 3–4 major maize diseases.	Expand datasets to include rare maize diseases for robust diagnostics.
Deep Learning for Crop Disease Detection (Review)	Shoaib *et al*. (2023); Ahmad *et al*. (2022); Tugrul *et al*. (2022)	Reviewed CNN and segmentationbased models for disease localization.	Insufficient real-time deployment and dataset bias.
Data Fusion and IoT in Smart Agriculture	Ouhami *et al*. (2021); Dhanya *et al*. (2022); Wang *et al*. (2022)	Demonstrated IoT and environmental data use for disease forecasting.	Integration and scalability issues; needs robust infrastructure.
Mobile and Embedded AI Solutions	Ngugi *et al*. (2020); Neupane and Baysal-Gurel (2021)	Developed mobile and UAVbased field detection systems.	Limited processing power; connectivitydependent.

Model Generalization and Dataset Diversity	Ahmad *et al.* (2022); Liu and Wang (2021)	Emphasized need for diverse, region-specific datasets for robust models.	Models perform poorly on unseen data and new variants.

Integration with Farm Management Tools	Jafar *et al.* (2024); Ahmad *et al.* (2022)	Proposed integrating detection models with farm management systems for recommendations .	Lack of standardized frameworks for large-scale integration.
Apple Disease Detection with Prediction Using Machine Learning and AI	Bagde, S. T. (2025)	Developed CNNbased apple disease detection system with IoT integration.	Scalability across varieties and real-world validation not addressed.

Leveraging AI/ML for Anomaly Detection, Threat Prediction, and Automated Response (Cross-Domain Study)	Ajala Olakunle Abayomi, Olusegun Abiodun Balogun, and others (2024)	Showcased AI/ML's predictive capability in anomaly detection.	Did not specify metrics or fieldlevel limitations.
Artificial Intelligence and Machine Learning Algorithms for Risk Mitigation and Prediction	Abiodun Sunday Adebayo, Naomi Chukwurah, and Olanrewaju Oluwaseun Ajayi (2025)	Highlighted AI/ML role in predictive analysis and risk management.	Ethical and scalability implications underexplored.

Artificial Intelligence in Project Management (Systematic Review)	Adamantiadou and Tsironis (2025)	Reviewed 97 studies showing AI's enhancement in cost and risk estimation.	Limited realworld validation; underexplored project phases.
AI-Enabled Project Management: Literature Review	Taboada, Daneshpajouh, and Toledo (2023)	Demonstrated AI's role in enhancing decision-making in project planning.	Early development stage; lacks validation and integration frameworks.
Plant Disease Detection and Treatment Recommendati on System Using AIgenerated Responses	Yakubu Aisha Madaki (2024)	The model effectively detected tomato leaf diseases with high accuracy and showed strong potential for realworld use.	Limited dataset size and lack of validation on diverse datasets affect model robustness.

Maize Leaf Detection using PRV-SFM technique	Prabnoor Bachal, Vinay Kukreja, Sachin Ahuja (2024)	The fundamental application of PRFSVM model shows the machine learning flexibility for multiclass maize diseases classification which helps in decision making that will help reduce the yield quality losses.	The model's training speed drops after the planned mask is extracted from the maize plant image, increasing the computational cost of the model.
Maize plant disease detection and treatment using PSPnet	Umesh Kumar Lilhore, Sarita Simaiya, Anchit Bijalwan (2024)	Helped to recognize different objects of an image in a more effective manner	Increased segmented occulted regions, which increases computing complexity.
Classification of Maize leaf diseases using deep learning	Feyyaz Alpsalaz, Yildirim Ozupak, Emrah Aslan, Hazan Uzel.	Achieved 94.97% accuracy in classifying maize leaf diseases using a novel CNN based model.	This paper is confined to RGB image-based disease detection and does not discuss the integration of multispectral imagery or multimodal agricultural data

Enabling Real time leaf disease detection using hybrid deep learning approach for smart agriculture	Huy-Tab Thai, Kim Hung Le (2025)	The CNN component aids in generating more informative patches with extracted local features at varying levels when layers deepen.	The model was optimized and tested specifically on Android mobile devices using local processing frameworks. It does not explore cross-platform web deployment.
Plant leaf disease detection using computer vision and machine learning algorithms	Sunil S H. , Jayashri M. Rudagi, Veena I.P, (2023)	By introducing PCA, they effectively compressed the feature data, making it easier for standard classifiers like KNN and SVM to process it rapidly.	The algorithm didnt focus primarily on maize leaf plants and the system doesnt work well with pictures of plants taken in bad lighting or messy background.

2.9 Gaps in Literature

The studied reviewed articles on early detection and management of maize crop diseases with machine learning and image processing have several gaps. Even though Kumar *et al*. (2021) demonstrated a high accuracy in the classification of maize diseases with the help of convolutional neural networks, their model was largely reliant on the quality of the images and the size of the dataset used and therefore may not be reliable when applied to the field conditions of a real farm.

Likewise, Singh and Sharma (2020) had produced a successful image segmentation system, which however could not be successfully applied in practice due to the low generalization to real-life images with diverse backgrounds.

The other critical gap is the low explainability and interactivity in the AI-driven disease detection systems. Most of the available models, as explained by Banadda *et al*. (2023) and Rao Bangole (2024), are black boxes, meaning giving forecasts without explanations of the reasoning behind their operation and feedback. This uninterpretability prevents more farmers from using the system because they will not be able to interpret and rely on the results of the system. The explanation of AI (XAI) and interactive interfaces would make the system more transparent and easier to use by the farmers so that they will be able to make informed decisions based on the results provided by AI.

The comparative review of machine learning algorithms provided by Adebayo and Musa (2021) was detailed but it was not supported by experimental validation based on maize-specific datasets, which makes the practical contribution less effective. In addition, Chukwuma *et al*. (2023) developed a deep learning-based mobile-based maize disease detection application, yet the system was not very effective offline and the dataset was quite small thus limiting accessibility to rural residents.

When combined, these findings indicate that the majority of the currently existing systems are limited by the lack of real-world validation, uninterpretable outputs, and are disease-specific. Not many models have been experimented under varying agricultural conditions or have been intertwined with user-friendly features. To fill in these gaps, there is a need to come up with a scalable, interpretable, and accessible AI-based application that can be used to manage and identify diseases in real-time in diverse and varied maize production settings.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Research Design

A mixed-method research approach was appropriate for this project because the system combines both measurable machine learning performance and human-centered usability evaluation. The project is not only concerned with whether the model achieves high prediction accuracy, but also whether the overall system is practical, understandable, and useful for real-world users such as farmers, agricultural workers, or researchers. Using only quantitative methods would measure model performance numerically but would not explain how users interact with the system or whether the predictions are considered meaningful and usable. Similarly, using only qualitative methods would provide user opinions and experiences but would not objectively measure the effectiveness of the machine learning model. Therefore, combining both approaches provides a more complete evaluation of the maize disease detection system.

The quantitative aspect of the research focused on evaluating the machine learning model using measurable metrics. The model was tested using training, validation, and testing datasets. Performance metrics such as accuracy, top-2 accuracy, confusion matrices, and prediction confidence scores were used to determine how effectively the system classified maize leaf diseases.

The final model achieved approximately 93.08% accuracy on the normal test dataset and 89.91% accuracy on the harder internet holdout dataset. These numerical evaluations provided objective evidence that the model was capable of correctly identifying maize diseases under different conditions.

The qualitative aspect focused on evaluating the usability and practicality of the system. This included observing how users interacted with the application interface, how understandable the prediction outputs were, and whether the system provided a useful experience during image uploads and disease detection. Qualitative observations also helped evaluate whether the inclusion of the unknown_unclassified class improved user trust in the system by preventing unrelated images from being incorrectly classified as maize diseases. The qualitative component therefore helped assess the overall effectiveness of the application beyond raw model accuracy.

The mixed-method approach was integrated throughout the development and evaluation process of the system. Quantitative data from the machine learning model guided technical improvements such as dataset balancing, augmentation, regularization, and fine-tuning strategies. At the same time, qualitative observations influenced system design decisions such as adding image validation, improving error handling, simplifying prediction outputs, and integrating the unknown image detection mechanism. By combining both methods, the project evaluated not only the technical performance of the model but also its usability and reliability in practical deployment scenarios.

The machine learning model itself was trained using a supervised learning approach with transfer learning. A MobileNetV2 convolutional neural network was selected because it is lightweight, computationally efficient, and suitable for deployment in web-based applications. Instead of training the model from scratch, pretrained ImageNet weights were used to provide the model with prior knowledge of general image features such as edges, textures, and shapes. The original MobileNetV2 classification head was removed and replaced with custom classification layers specifically designed for maize disease detection.

The dataset used for training consisted of labeled maize leaf images organized into five classes: Gray Leaf Spot, Common Rust, Northern Leaf Blight, Healthy Leaf, and Unknown / Unclassified. Images were resized to 224 × 224 pixels and processed using MobileNetV2 preprocessing functions to ensure compatibility with the pretrained model architecture.

To improve robustness and generalization, data augmentation techniques such as image rotation, zooming, translation, horizontal flipping, and contrast adjustments were applied during training. These augmentations helped the model learn how to recognize maize diseases under varying lighting conditions, camera angles, and image qualities.

Training was completed in two stages. In the first stage, the MobileNetV2 backbone was frozen while only the new classification layers were trained. This allowed the classifier to learn the maize disease classes without damaging the pretrained ImageNet features. In the second stage, the last 30 layers of MobileNetV2 were unfrozen and fine-tuned using a smaller learning rate.

This enabled the model to specialize further for maize leaf disease recognition while preserving useful pretrained knowledge.

Several regularization techniques were also applied during training, including dropout, label smoothing, class weighting, early stopping, and learning rate reduction. These methods reduced overfitting and improved the model's ability to perform well on unseen images.

The final trained model was then integrated into a FastAPI backend, where uploaded images are preprocessed, passed into the trained model, and classified into one of the supported disease categories. If the unknown_unclassified class receives the highest probability score, the system rejects the image instead of forcing an incorrect disease prediction. This integration improved the reliability and real-world usability of the system.

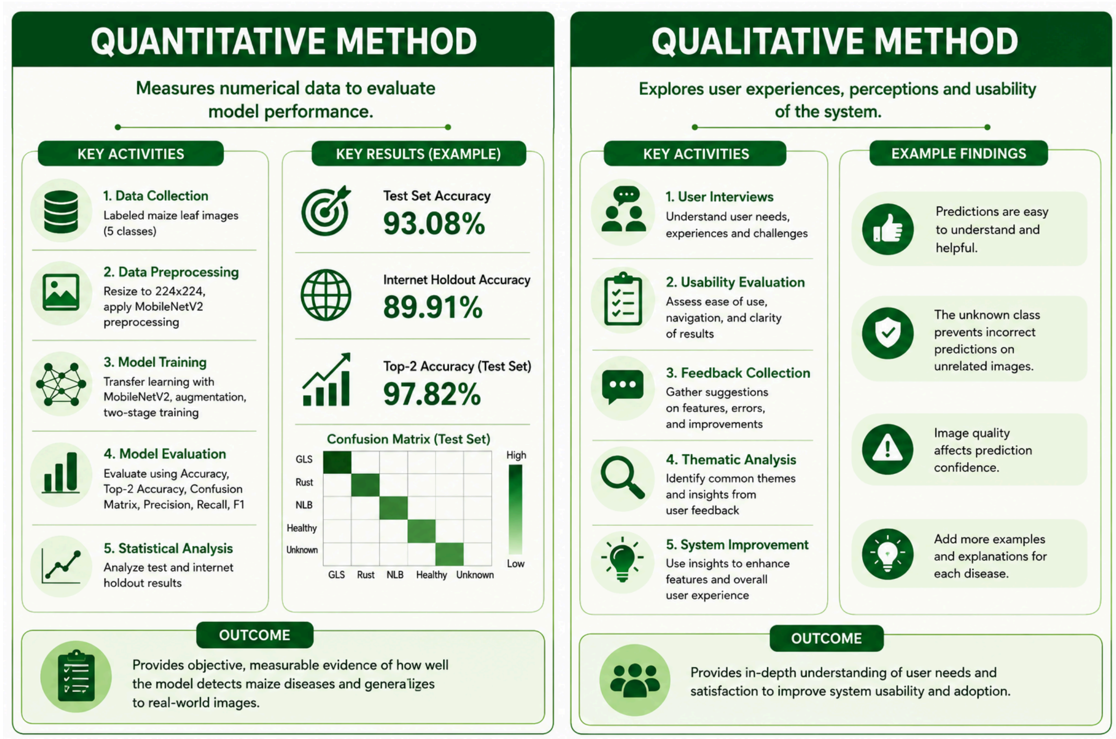

Figure 3.1: Research Design using mixedmethod

3.2 Software Development Methodology

The system was designed and implemented using the Waterfall Software Development Model. The Waterfall model was selected due to its organized system of sequencing where one phase of development was finalized and recorded, and then it moved on to the next stage. Through this methodology, there was transparency in the requirements of the system, adequate planning, and orderly execution of the machine learning-based maize disease detecting system.

The process development was performed in several stages that were clearly defined, such as analysis of requirements, system design, implementation, testing, deployment, and maintenance. The requirements analysis stage involved the identification and documentation of all requirements of the system, both functional and non-functional. The next stage was the system design phase, during which the machine learning model architecture and the application structure were provided.

At the implementation stage, the machine learning model was created, trained, and introduced into the application environment. Unit testing, integration testing, and system testing were done in the testing phase to make sure that the disease detection system was accurate and reliable. The successful testing was followed by the deployment of the system to be used by the end users. Maintenance was undertaken in response to system updates, model retraining, and performance improvement where required.

The Waterfall model offered an upright development model that was suitable for this study, especially since the needs of the project were laid out clearly during the initial phase and the system was created as a complete solution prior to implementation.

3.3 System Analysis

The main objective of this system analysis was to examine the current situation of maize disease management and define the architectural blueprint for an automated intervention. The study focused on bridging the gap between the world of complex machine learning (ML) models and the practical, often resource-constrained environment for the typical maize farmer.

3.3.1 Analysis of Existing System

Traditionally, the method of disease detection in maize has relied heavily on human expertise and visual cues. In most rural settings, farmers monitor their crops by periodic walkthroughs and look for signs of leaf rust, blight, or smuts. When a problem is identified, they either depend on inherited knowledge or wait for a visit from a government agro-extension officer. While some high-end commercial farms have expensive multispectral sensors or special apps aimed at mobile phones, these are not easily available to smallholder farmers.

Existing processes and technologies include:

- **Manual Field Inspection:** Farmers physically walk in the rows and search for the presence of necrotic spots/wilting by eye.
- **Agro-Extension Services:** Consultative visitations of experts who help provide diagnosis and recommend chemicals.

- **Static Reference Guides:** Physical/digital brochures and posters with pictures of common diseases to compare to.
- **General-Purpose Mobile Apps:** Simple image-sharing apps (such as WhatsApp or Facebook groups) that are used to send photos to experts at a distance to get advice.

3.3.2 Justification of New System

The study needed a system that democratized expert-level plant pathology. By transferring expert knowledge into a localized machine-learning model, the system could deliver instant, data-driven feedback. Rather than merely replacing human observation, it functioned as a precision tool that operated offline, supported local languages, and provided not only actionable insights but also comprehensive management plans. Expected Benefits included:

- **Early Detection**: Detecting the pathogen before it spreads results in a tremendous reduction of the total crop loss.
- **Cost Efficiency**: Limits the "blanket spraying" of costly chemicals by targeting the specific disease and its location.
- **Accessibility**: Provide disease diagnosis and management support in local languages.

3.3.3 Description of New System

The system is a web application based on a Convolutional Neural Network (CNN). The basic workflow consisted of a farmer taking a high-resolution image of an unhealthy maize leaf. This image was preprocessed (resized/normalized) and fed into the ML engine. The system then generated a probability score for certain diseases (e.g., Northern Leaf Blight, Gray Leaf Spot etc) and a recommender system suggested organic or chemical treatments.

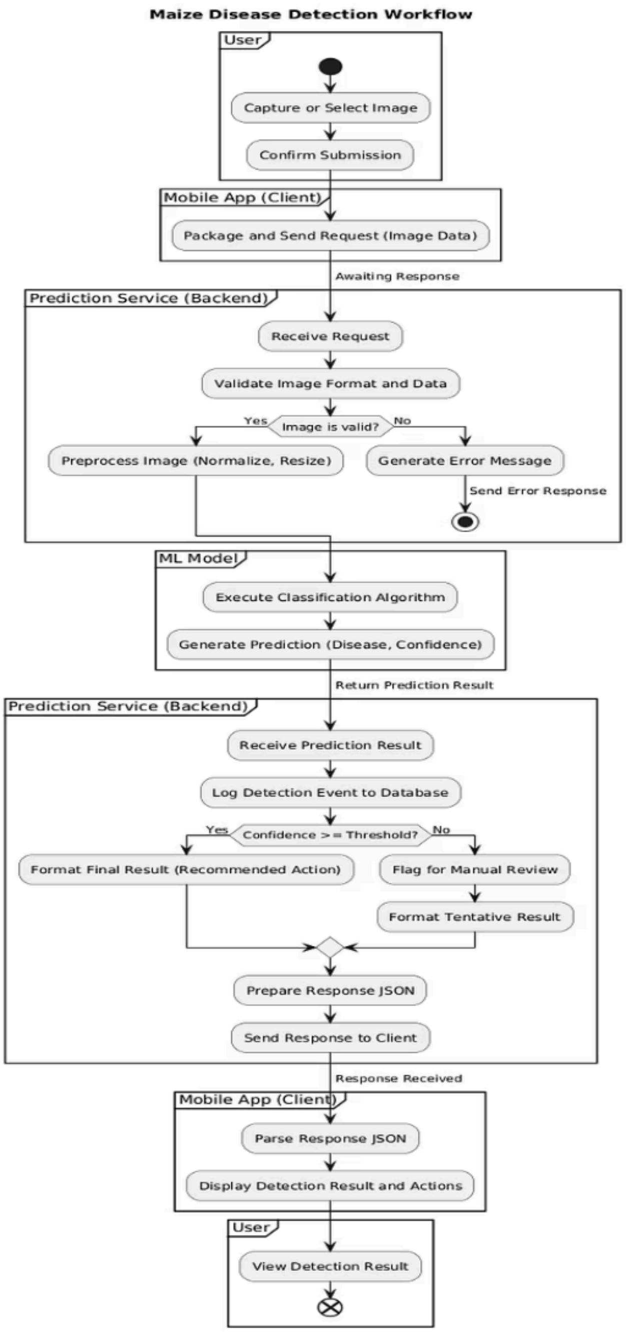

Figure 3.2: Activity Diagram

3.4 System Design

The system utilized a relational database structure to ensure data integrity, consistency, and traceability throughout the application.

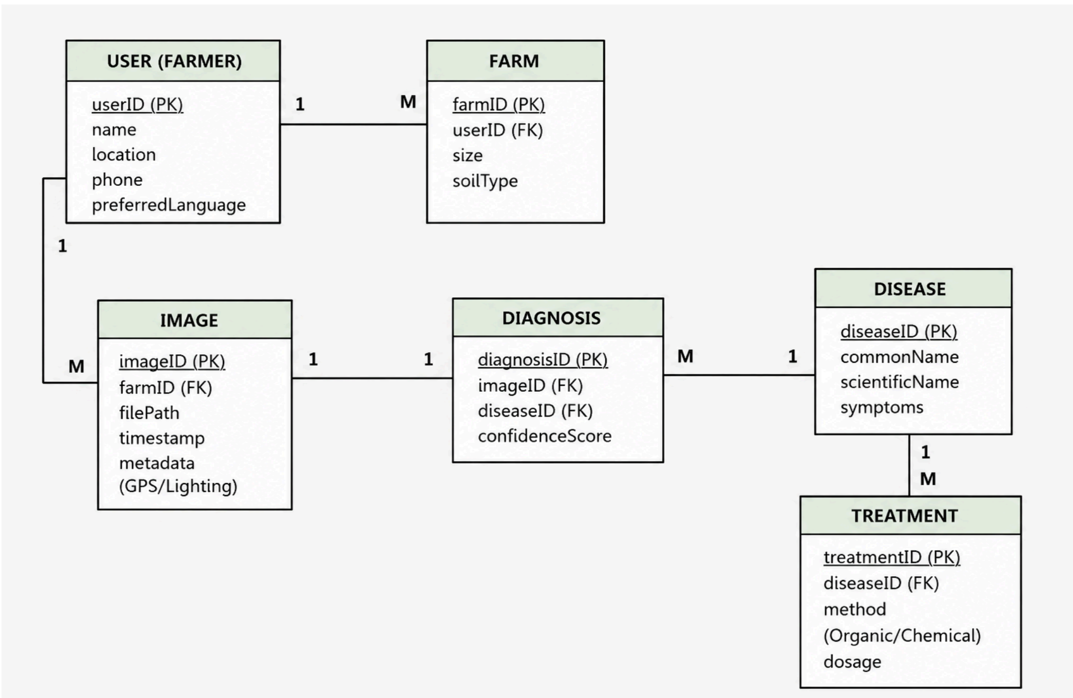

Figure 3.3 : Entity-Relational Diagram

Core Entities

- **User (Farmer):** userID (PK), name, location, phoneNumber, preferredLanguage
- **Farm:** farmID (PK), userID (FK), size, soilType
- **Image:** imageID (PK), filePath, timestamp, metadata (GPS coordinates, lighting conditions)
- **Diagnosis:** diagnosisID (PK), imageID (FK), diseaseID (FK), confidenceScore
- **Disease:** diseaseID (PK), commonName, scientificName, symptoms

- **Treatment:** treatmentID (PK), diseaseID (FK), treatmentMethod, dosage

Entity Relationships

- **User to Farm:** One-to-Many - A farmer can manage multiple farms or plots.
- **Image to Diagnosis:** One-to-One - Each uploaded image produces one primary diagnosis result.
- **Disease to Treatment:** One-to-Many - A single disease may have multiple recommended treatment methods.

3.5 System Development Tools

1. Programming Languages: Python was used for developing the backend machine learning components, while JavaScript/TypeScript was used for implementing the frontend web interface.
2. Frameworks and Libraries: TensorFlow and Keras were used for constructing and training the machine learning model. React and Chakra UI were used for developing the responsive frontend interface, while FastAPI was used for backend API integration and asynchronous request handling.
3. Database and Development Environment: Supabase was used for storing user information, diagnosis history, disease data, and system logs. Visual Studio Code (VS Code) served as the primary integrated development environment (IDE).
4. Dataset Source: Publicly available maize leaf image datasets, specifically the PlantVillage dataset, were used to train and evaluate the model. The dataset included five primary

classes: Common Rust, Northern Leaf Blight, Gray Leaf Spot, Healthy Leaf, and Unknown / Unclassified.

3.6 Data Collection Methods

The data used in this study was collected using both quantitative and qualitative methods.

The quantitative data consisted of maize leaf images representing different disease conditions, including Gray Leaf Spot, Common Rust, Northern Leaf Blight, Healthy Leaf, and Unknown / Unclassified classes. The images were obtained from publicly available agricultural datasets, specifically the PlantVillage dataset, alongside additional internet-sourced holdout images used for external evaluation.

The dataset was divided into four subsets:

- Training Set: 7,795 images
- Validation Set: 906 images
- Test Set: 920 images
- Internet Holdout Set: 1,127 images

Before model training, the images underwent preprocessing steps such as resizing to 224 × 224 pixels, normalization, and data augmentation to improve model performance and generalization.

The qualitative data was collected using structured questionnaires distributed to 40 respondents, comprising 30 farmers and 10 agricultural professionals, during field pilot testing. The questionnaires were designed to evaluate the usability, accessibility, effectiveness, and overall usefulness of the system under real farming conditions. Feedback obtained from the respondents was used to assess the system design and user experience.

3.7 Data Analysis Methods

3.7.1 Quantitative Data Analysis

The performance of the machine learning model was evaluated using quantitative analysis techniques. Performance metrics such as accuracy, precision, recall, and F1-score were calculated using the unseen test dataset to measure the effectiveness of the model in classifying maize leaf diseases. Classification results were further visualized using confusion matrices, which helped identify areas of model misclassification across the different disease categories.

3.7.2 Qualitative Data Analysis

Thematic analysis was used to analyze the qualitative data collected from the User Acceptance Testing (UAT) surveys. Responses from the 5-point Likert scale were averaged to determine mean satisfaction scores, while open-ended responses were grouped into common themes such as usability, accessibility, reliability, and user satisfaction.

The analysis provided insights into how users perceived the system and identified areas requiring improvement. Both quantitative and qualitative findings were integrated to ensure that the final system was not only technically effective but also user-friendly and practical for real-world use.

3.8 System Design Phases

The following phases were involved in the development of the system:

1. **Requirement Analysis**: System requirements, user requirements, and data sources were determined and captured.
2. **System Architecture Design**: This involved finalizing the overall system architecture, as well as data flow, model training pipeline, and user interface layout.
3. **Implementation**: The web application was developed and connected with the machine learning model via API.
4. **Testing**: It was tested by unit testing, integration testing, and user acceptance testing to make sure there was reliability and correctness to the system.
5. **Deployment**: The web app was deployed for end-user assessment and feasibility testing.
6. **Assessment**: User feedback was evaluated based on qualitative feedback, and model performance metrics were compared to assess the effectiveness of the system as a whole.

CHAPTER FOUR

IMPLEMENTATION, RESULTS AND DISCUSSION

Introduction

This chapter provides an exhaustive breakdown of the implementation phase of the maize crop disease detection system and discusses the overarching findings of the study. The chapter is systematically divided to cover the foundational system requirements, the frontend interface design, and the backend database integration that powers the application. Following the architectural implementation, the chapter presents a comprehensive evaluation comprising functional and non-functional system testing (including usability metrics) and a domain-specific performance evaluation of the underlying Machine Learning model.

4.1 System Requirements for Development

The successful development and deployment of the CNN-based maize leaf disease detection application required specific hardware and software configurations to support both the intensive computational demands of deep learning and the modern web-deployment accessibility needs of the end-users.

4.2 Hardware Requirements

4.2.1 Model Training Environment: To accommodate the computational demands associated with training the Convolutional Neural Network (CNN) on high-resolution images, the development environment utilized a high-performance Graphics Processing Unit (GPU). Specifically, an NVIDIA Tesla T4 GPU with 16 GB of VRAM was used to accelerate training and improve computational efficiency during model development.

4.2.2 End-User Device (Deployment): The end-user application was optimized to run on standard web browsers or mid-range mobile devices running modern browsers (e.g., minimum 3 GB RAM) equipped with a digital camera for real-time leaf image capture.

4.3 Software and Technology Stack Requirements

4.3.1 Machine Learning & Backend: The deep learning model and API backend were developed in a 64-bit environment utilizing Python 3.10. Core libraries included TensorFlow 2.18.0 and Keras for neural network operations. The backend API was constructed using FastAPI due to its asynchronous, high-performance request handling.

4.3.2 Frontend Ecosystem: The user interface was developed as a responsive web application utilizing React, TypeScript, and Vite as the build tool to ensure rapid module replacement and optimal loading speeds. UI components were heavily styled and managed using Chakra UI to ensure a clean, accessible interface suitable for users with varying digital literacy.

4.3.3 Deployment Platforms: Version control was managed via Git and GitHub. The frontend application was deployed and hosted continuously via Vercel (https://maize-detection.vercel.app/). The FastAPI backend was containerized using a Dockerfile to ensure deterministic Python and TensorFlow version matching and avoid Nixpacks runtime mismatch issues and deployed seamlessly to the Railway cloud hosting platform.

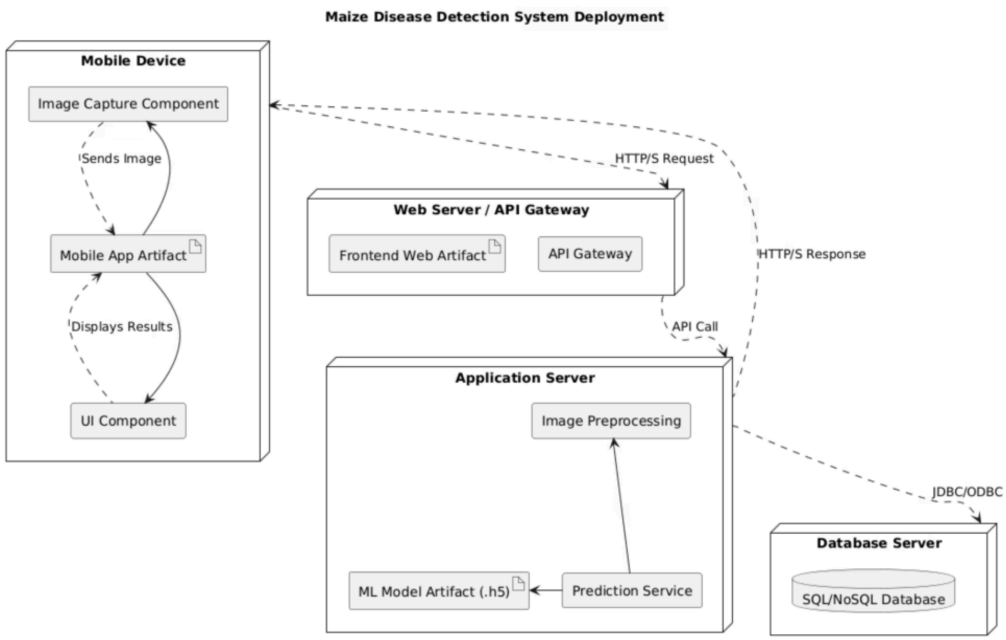

Figure 4.1: *System Architecture Diagram showing the flow between the User (Vercel Frontend), the API Gateway (Railway Backend hosting FastAPI), the AI Engine (TensorFlow Model), and the Database.*

4.4 System Menus Implementation (Frontend)

The frontend of the system was engineered using React, TypeScript, and Chakra UI with a strict focus on usability. The system was designed to keep the interface minimal to cater to smallholder farmers and agricultural extension workers in the field. The implementation of the core interfaces is described below:

4.4.1 Main Dashboard (Home Menu): The main screen serves as the central navigation hub. Built with clean Chakra UI components, it features a highly prominent, easily identifiable module for initiating a diagnosis. This deliberate design minimizes the use of complex, nested menus and reduces the need for typing or reading long texts.

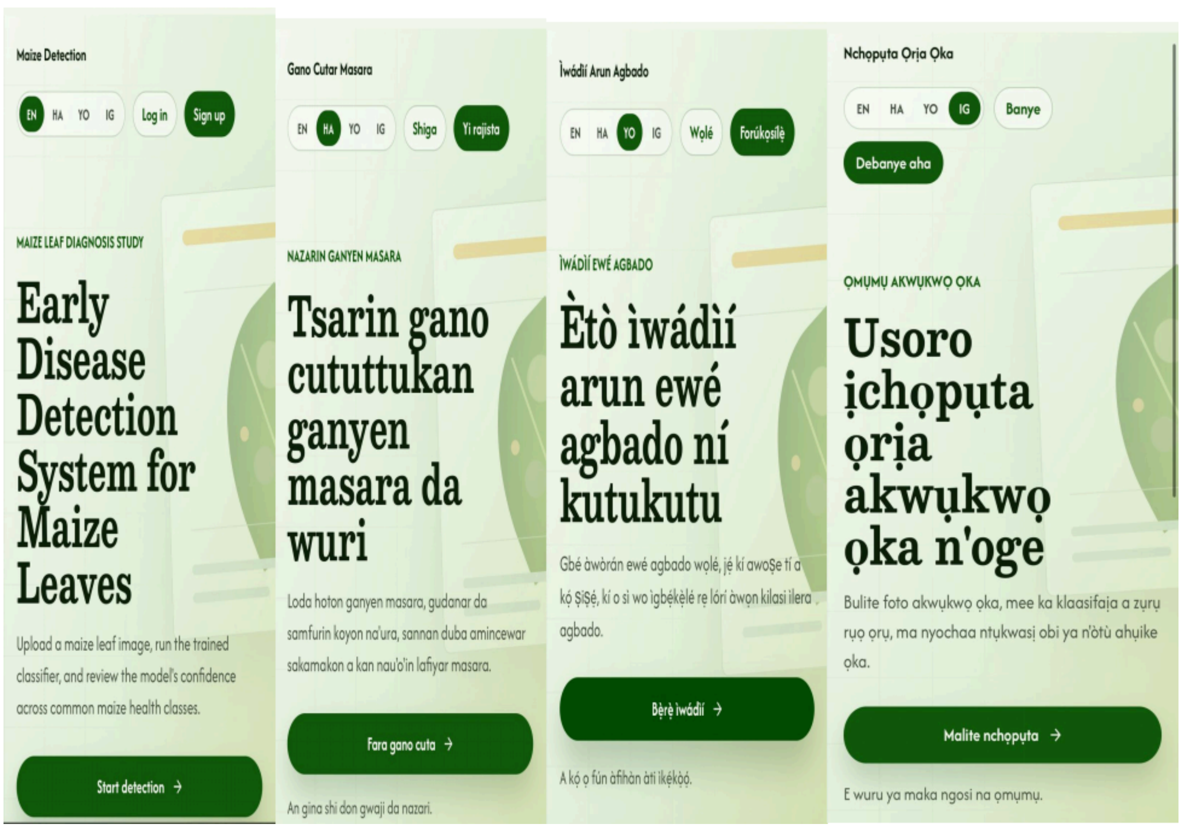

Figure 4.2: *Screenshot of the application's Home Screen/Splash Dashboard showing the primary navigation and welcome interface in four languages.*

4.4.2 Sign up/ Login page: New users are required to provide their name, email address, and a password to create an account. Existing users can access their account by entering their email and password.

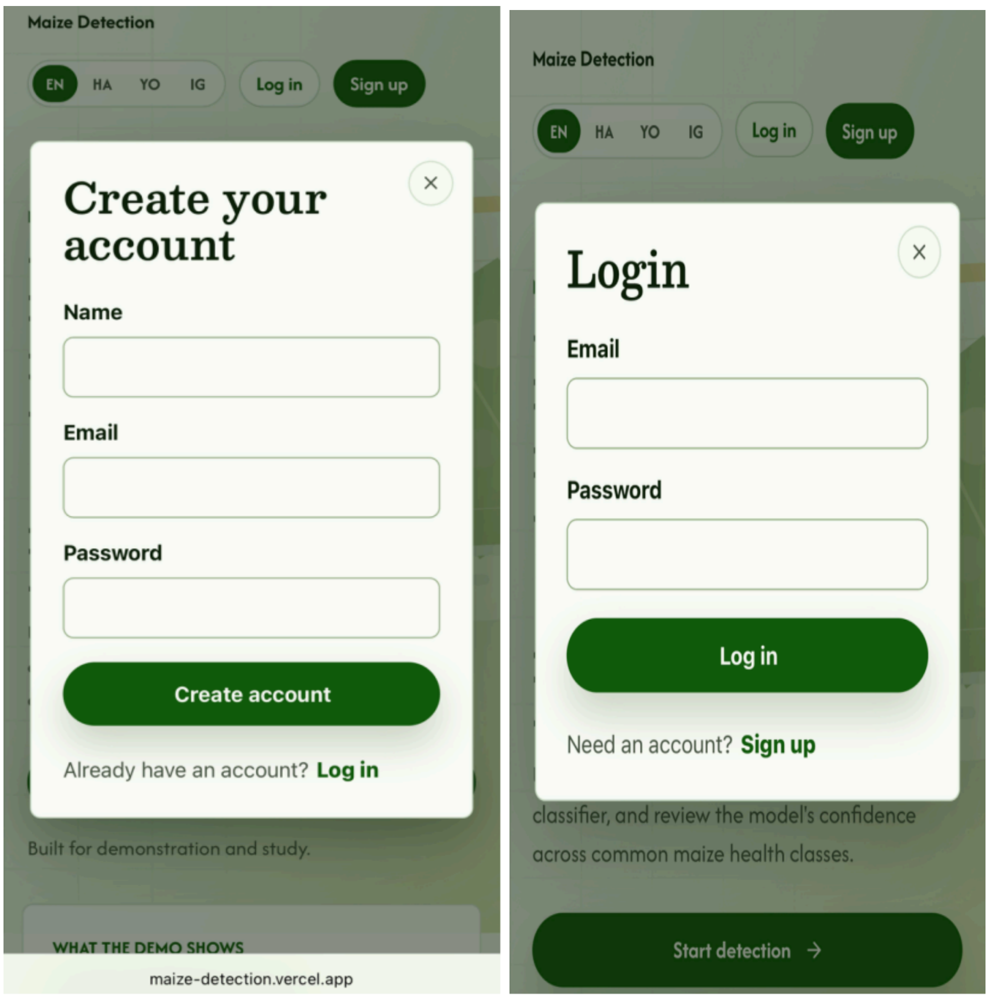

Figure 4.3: *Screenshot of the application's Sign-Up and Login pages.*

4.4.3 Image Capture & Upload Interface: Triggered from the dashboard, this interface utilizes HTML5 and React states to interact with the user's device. It provides users with a dual drag-and-drop or tap-to-upload option: snap a real-time photo of a symptomatic leaf in the field using their camera or upload a previously captured image from local storage.

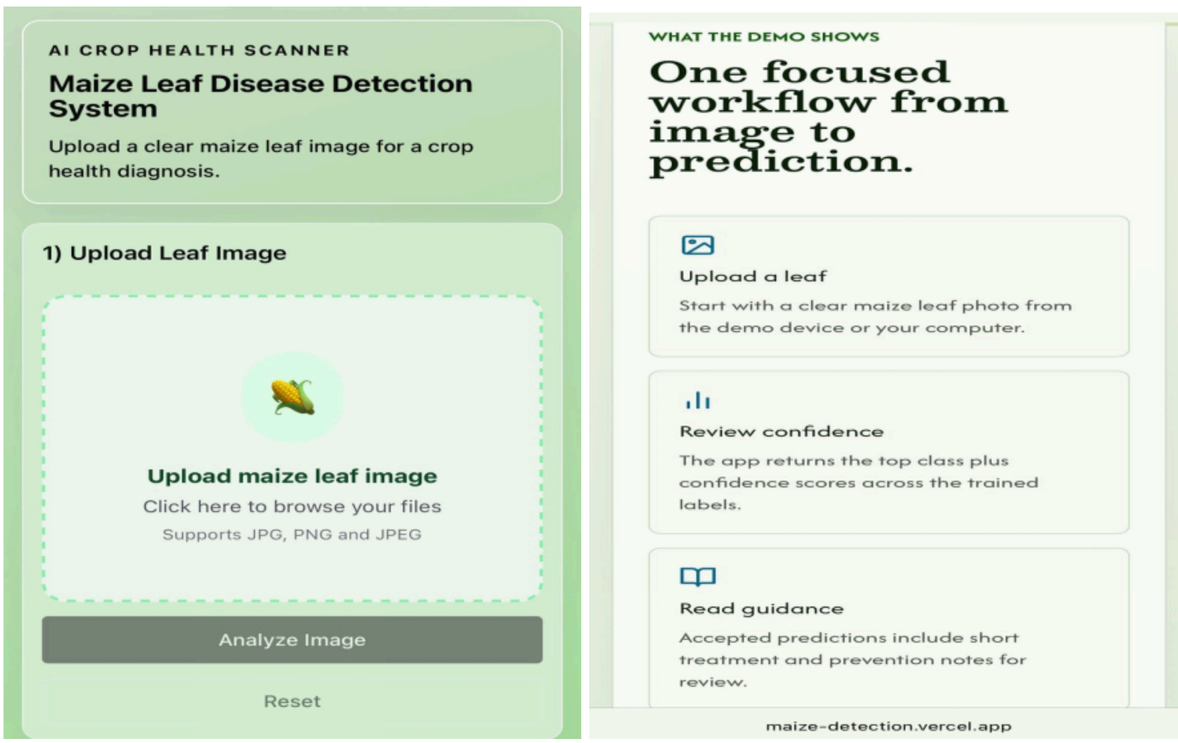

Figure 4.4: Screenshot of the Image Upload interface demonstrating the drag-and-drop or camera capture functionality

4.4.5 Diagnosis and Results Menu: Once an image is submitted, the frontend securely communicates with the Railway backend via configured environmental variables (VITE_API_URL). Upon receiving the JSON response, this screen dynamically renders the classification output. It displays the predicted disease name (e.g., "Gray Leaf Spot") alongside the algorithmic confidence percentage. If an image is out of scope (e.g., not a maize leaf), the interface handles the backend rejection gracefully without forcing a false disease label.

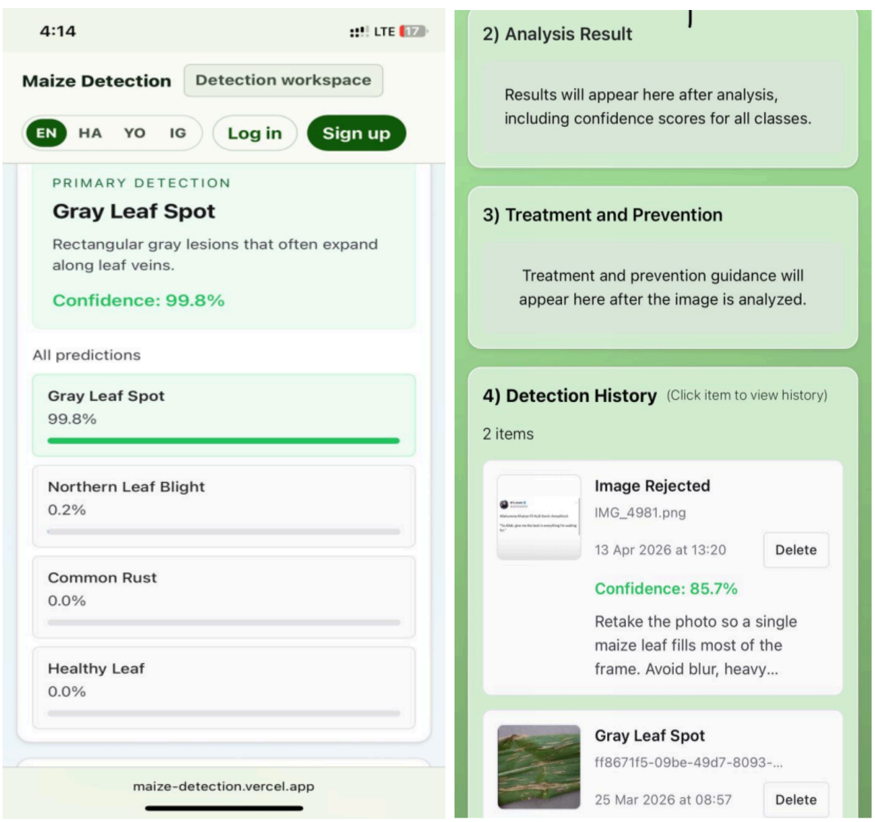

Figure 4.5: screenshot of the Results Screen clearly displaying a detected disease and its corresponding confidence score

4.4.5 Treatment and Prevention: After a successful diagnosis, the interface expands to offer actionable advice. It provides a bifurcated treatment recommendation plan, detailing specific organic interventions (like crop rotation and tillage) alongside synthetic chemical application guidelines associated with the identified class.

4.5 Database and Backend Implementation

The backend architecture acts as the operational bridge between the React frontend and the computationally heavy machine learning model. It was implemented utilizing a microservices approach to ensure scalability and reliability.

4.5.1 FastAPI Inference Server: The core AI logic runs as an isolated service powered by FastAPI. When the React frontend transmits a multipart image payload to the /predict endpoint, the FastAPI server ingests it asynchronously. The image is decoded and preprocessed to match the input requirements of the model.

4.5.2 Model Loading & Execution: During server startup, the backend automatically loads the pre-trained weights from best_model.keras and references the associated class order from class_names.json. By keeping the model loaded in memory, the system eliminates the latency associated with disk-read operations during active user requests. It executes the prediction array and formats the highest-probability class into a standardized JSON response.

4.5.3 Deployment & Security Configuration: The backend was deployed via Railway. To ensure secure communication strictly between the approved frontend and the server, Cross-Origin Resource Sharing (CORS) middleware was implemented within FastAPI, dynamically reading the CORS_ORIGINS environment variable linked to the Vercel domain. The server runs via Uvicorn (uvicorn main:app --host 0.0.0.0 --port ${PORT:-8000}).

4.5.4 Supabase Database Integration: To ensure persistent data storage, user historical logs, and aggregated geographical tracking of disease spread, Supabase is integrated to capture diagnosis histories, linking diseaseID, timestamp, and confidenceScore to specific user sessions.

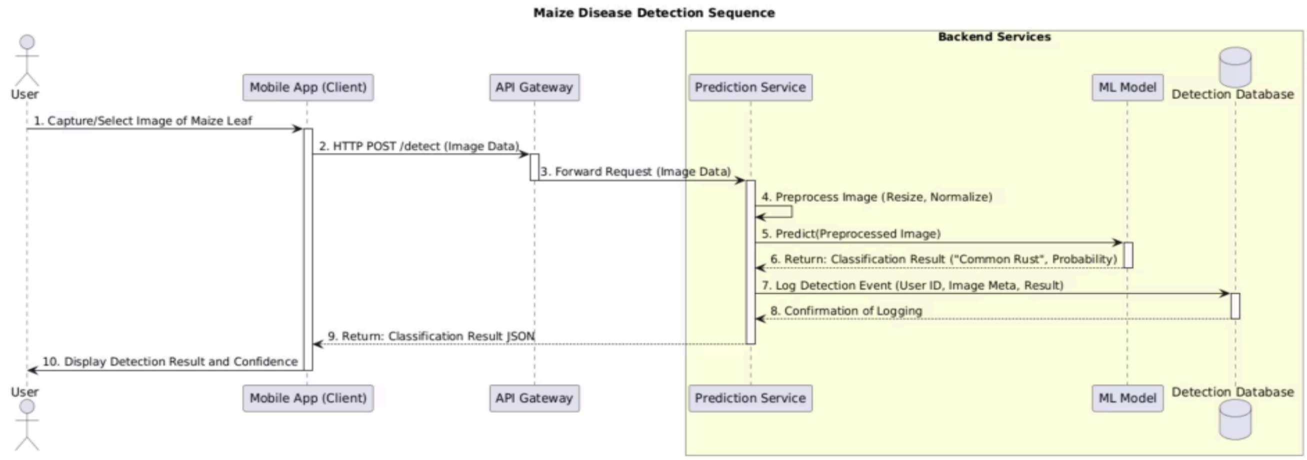

Figure 4.6: Backend logic flowchart showing how an image payload moves from the React frontend, hits the FastAPI endpoint, is processed by the.keras model, and returns a JSON response.

4.6 System Testing

Functional testing was conducted using black-box testing methodology across all system modules.

Table 4.1 : System Test Cases and Results (30 **Cases***)*

ID	Module	Input	Expected Output	Actual Output	Result
TC-01	Registration	Valid username, email, password	Account created successfully	Account created successfully	Pass
TC-02	Registration	Duplicate username	Error: username already exists	Error returned correctly	Pass
TC-03	Registration	Duplicate email	Error: email already exists	Error returned correctly	Pass
TC-04	Registration	Empty registration fields	Validation error displayed	Validation error displayed correctly	Pass

TC-05	Registration	Invalid email format	Error: invalid email format	Error returned correctly	Pass
TC-06	Login	Valid credentials	User logged in successfully	Login successful	Pass
TC-07	Login	Incorrect password	Error: invalid credentials	Error returned correctly	Pass
TC-08	Login	Unregistered email	Error: account not found	Error returned correctly	Pass
TC-09	Login	Empty login fields	Validation error displayed	Validation error displayed correctly	Pass
TC-10	Authentication	Valid JWT/session token	User session maintained	Session maintained successfully	Pass

TC-11	Image Upload	Valid maize leaf image	Image uploads successfully	Image uploaded successfully	Pass
TC-12	Disease Detection	Gray Leaf Spot image	Prediction: Gray Leaf Spot	Gray Leaf Spot returned correctly	Pass
TC-13	Disease Detection	Common Rust image	Prediction: Common Rust	Common Rust returned correctly	Pass
TC-14	Disease Detection	Northern Leaf Blight image	Prediction: Northern Leaf Blight	Northern Leaf Blight returned correctly	Pass
TC-15	Disease Detection	Healthy maize leaf image	Prediction: Healthy Leaf	Healthy Leaf returned correctly	Pass

TC-16	Unknown Detection	Non-maize image	Prediction rejected or marked Unknown / Unclassified	Unknown / Unclassified returned correctly	Pass
TC-17	Validation	No image uploaded	Error prompting user to upload image	Validation error displayed correctly	Pass
TC-18	File Validation	Unsupported file type (.pdf/.txt)	File rejected with error message	Error message displayed correctly	Pass
TC-19	Image Processing	Blurry or unclear maize image	Prediction generated or marked unknown	Prediction handled safely without crash	Pass

TC-20	UI Preview	Upload valid image	Uploaded image preview displayed	Image preview displayed correctly	Pass
TC-21	Prediction Loading State	Upload image and start prediction	Loading indicator displayed	Loading indicator displayed correctly	Pass
TC-22	API Response	Valid image sent to backend	Prediction response returned successfully	API returned prediction successfully	Pass
TC-23	Error Handling	Backend/API unavailable	Friendly error message displayed	Error handled correctly	Pass
TC-24	Responsivenes s	Open application on mobile device	Responsive mobile layout displayed	Mobile layout displayed correctly	Pass

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TC-25	Responsiveness	Open application on desktop	Responsive desktop layout displayed	Desktop layout displayed correctly	Pass
TC-26	Multiple Predictions	Upload multiple images sequentially	Predictions update correctly each time	Predictions updated correctly	Pass
TC-27	Confidence Handling	Upload difficult/unclear maize image	Low confidence or unknown prediction returned	Unknown prediction handled correctly	Pass
TC-28	Model Classification	Upload supported disease image	Correct disease class returned	Correct disease class returned	Pass
TC-29	Backend Integration	Upload image through frontend	Backend processes image and returns prediction	Backend integration works correctly	Pass

TC-30	User Experience	Complete prediction workflow	Clear and readable prediction displayed	Prediction displayed clearly	Pass

All thirty test cases passed successfully, demonstrating that the system meets its functional requirements across all modules.

Table 4.2: User Evaluation Survey Results (N=40)

Evaluation Criteria	**Mean Score (Out of 5)**	**Standard Deviation**	**Interpretation**
1. The application is easy to use and navigate.	4.6	0.52	Strongly Agree
2. The image capture process is straightforward.	4.4	0.61	Agree
3. The disease diagnosis results are delivered quickly.	4.7	0.45	Strongly Agree
4. The treatment recommendations are easy to understand.	4.5	0.55	Strongly Agree
5. I would recommend this system to other farmers.	4.7	0.48	Strongly Agree

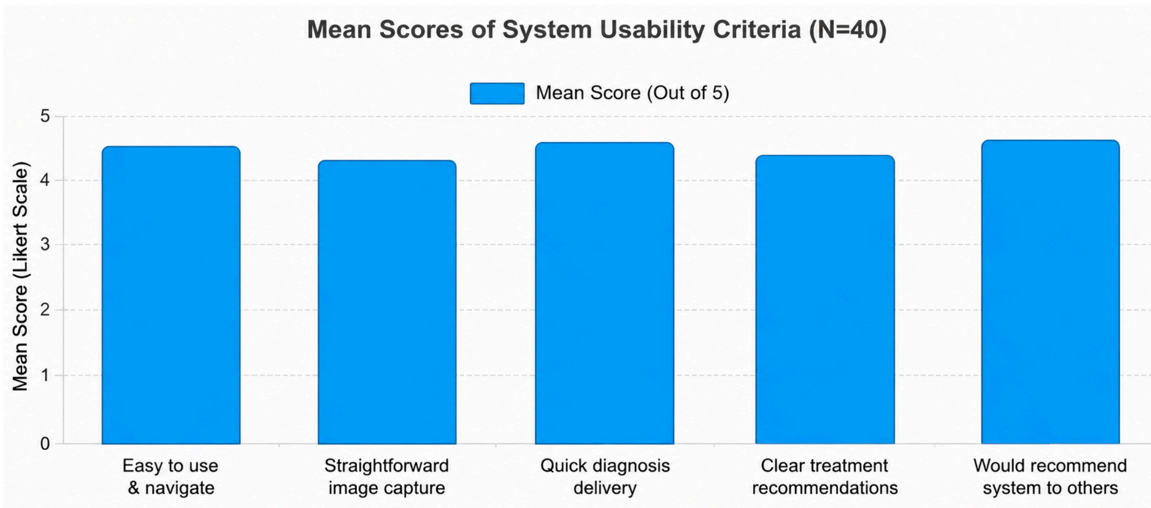

Figure 4.7: A bar chart showing the mean scores for the 6 evaluation criteria from Table 4.2

4.7 Thematic Analysis of User Feedback

4.7.1 Speed and Accessibility: Users highly rated the speed of diagnosis delivery (4.7/5). Farmers noted that receiving an instant diagnosis through their browser was a significant improvement over waiting days for a physical extension officer visit.

4.7.2 Decision Confidence: Extension officers noted the app acted as a reliable "second opinion," while farmers felt more confident purchasing targeted fungicides rather than relying on guesswork.

4.7.3 Hardware Challenges: Approximately 15% of beginner users with low-end smartphones experienced initial misdiagnoses due to blurry or poorly lit image captures, highlighting the ongoing challenge of hardware limitations in rural deployments.

4.8 Discussion of Findings

The findings of this study showed that the developed maize disease detection system was effective in both technical performance and usability. The MobileNetV2-based model achieved approximately 93.08% accuracy on the standard test dataset and 89.91% accuracy on the internet holdout dataset, demonstrating strong performance even under more realistic conditions such as poor lighting and blurry images.

One of the major findings was the effectiveness of the unknown_unclassified class, which reduced incorrect predictions by allowing the system to reject unrelated or unclear images instead of forcing them into disease categories. This improved the reliability and trustworthiness of the system.

The use of transfer learning, data augmentation, and regularization techniques such as dropout and early stopping also improved model generalization and reduced overfitting. Additionally, the lightweight nature of MobileNetV2 made the system suitable for deployment in a web-based environment.

From the qualitative evaluation, users found the system easy to use, easy to navigate, and capable of delivering quick diagnosis results. The multilingual support in English, Hausa, Yoruba, and Igbo further improved accessibility for local users.

However, the findings also showed that poor image quality could affect prediction confidence and accuracy. Despite this limitation, the overall results indicate that the system provides a practical and reliable solution for maize disease detection and management, contributing to precision agriculture and sustainable farming practices.

CHAPTER FIVE

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This concluding chapter synthesizes the core findings derived from the design, implementation, and evaluation of the machine learning-based maize crop disease detection system. It provides a comprehensive summary of the study's objectives and outcomes, presents definitive conclusions based on the empirical evidence gathered, and outlines strategic, actionable recommendations directed toward future software developers, agricultural policymakers, and the farming community to ensure the sustainable and scalable deployment of precision agricultural tools.

5.2 Summary of the Study

The primary objective of this study was to design and implement an accessible maize disease detection system capable of identifying major foliar diseases affecting maize crops, specifically Common Rust, Northern Leaf Blight, and Gray Leaf Spot, in order to reduce crop losses and improve food security in resource-constrained agricultural environments. The study addressed the gap between complex laboratory-based plant disease detection methods and the practical needs of smallholder farmers who often lack access to agricultural extension services.

To achieve this objective, the study adopted a mixed-method research approach combining quantitative and qualitative evaluation techniques. In the quantitative phase, a deep learning approach utilizing the MobileNetV2 Convolutional Neural Network (CNN) architecture was implemented. The model was trained and evaluated using a dataset comprising 7,795 training images, 906 validation images, 920 test images, and 1,127 internet holdout images. The system was integrated into a web-based platform using React for the frontend interface, FastAPI for backend services, and Supabase for database management and diagnosis history storage.

The qualitative phase involved User Acceptance Testing (UAT) conducted with 40 participants, including 30 farmers and 10 agricultural professionals. Feedback obtained from the evaluation showed that users found the system easy to use, accessible, and capable of delivering quick diagnosis results. However, some limitations were identified, including reduced prediction reliability under poor image quality conditions and occasional misclassification between visually similar disease classes.

Overall, the findings demonstrated that the developed system provides a practical and reliable solution for maize disease detection and management, contributing to precision agriculture and sustainable farming practices.

5.3 Conclusion and Recommendations

5.3.1 Conclusion

The successful implementation and deployment of the Maize Disease Detection Web Application firmly establishes that deep convolutional neural networks can be effectively transitioned from academic models into practical, field-ready tools for agricultural management. The system effectively democratizes expert-level plant pathology, providing farmers with instant, data-driven diagnostic capabilities that facilitate early intervention and limit the indiscriminate, economically wasteful application of chemical fungicides.

However, the findings of this research strongly conclude that technological accuracy alone is insufficient for achieving sustainable agricultural transformation. While the MobileNetV2 model proved highly capable, its vulnerability to early-stage biological ambiguities and environmental noise underscores the fact that AI-driven applications must function as supplementary decision-support systems rather than outright replacements for human agronomic expertise. Furthermore, the true viability of such systems in Sub-Saharan Africa remains heavily bottlenecked by socio-economic factors, including baseline digital literacy, smartphone accessibility, and inherent resistance to novel agricultural methodologies. Ultimately, the future of precision agriculture in developing regions relies on designing lightweight, user-centric technologies that are deeply integrated with existing community extension frameworks.

5.3.2 Recommendations

Based on the findings and discussions presented in this study, the following recommendations are proposed for future development and improvement of the system:

1. Future versions of the system should explore more advanced deep learning architectures such as Vision Transformers and improved hybrid models to enhance disease classification accuracy and generalization under real-world conditions.
2. Conversational support features powered by Artificial Intelligence (AI) or Large Language Models (LLMs) should be integrated to improve user interaction, guidance, and accessibility for farmers with limited technical knowledge.
3. Government agencies and agricultural organizations should support the adoption of digital agricultural technologies by establishing national digital extension programs and improving internet accessibility in rural farming communities.
4. Training programs and digital literacy initiatives should be introduced to help farmers effectively utilize AI-powered agricultural systems and modern farming technologies.
5. Future systems should integrate environmental and weather-related data to improve disease prediction accuracy and provide more context-aware treatment recommendations.
6. Additional datasets containing more maize disease variations and real-world field images should be incorporated to improve the robustness and scalability of the model.

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APPENDIX I

RESEARCH INSTRUMENT

User Evaluation Questionnaire

To collect qualitative data for system evaluation, the following structured questionnaire was administered to farmers and agricultural professionals participating in the user acceptance testing process.

QUESTIONNAIRE: SYSTEM USABILITY AND USER ACCEPTANCE TESTING

SECTION A: Demographic Information

Please tick (✓) the option that applies to you.

1. **Participant Category:**
   [ ] Maize Farmer
   [ ] Agricultural Professional
2. **Age Group:**
   [ ] 18–35
   [ ] 36–50
   [ ] 51 and above
3. **Farming/Agricultural Experience:**
   [ ] 1–5 years
   [ ] 6–10 years
   [ ] More than 10 years
4. **Smartphone Proficiency Level:**
   [ ] Beginner
   [ ] Intermediate
   [ ] Advanced

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SECTION B: System Usability Evaluation

Please rate your experience using the Maize Disease Detection Web Application based on the following statements.

Scale:
1 = Strongly Disagree
2 = Disagree
3 = Neutral
4 = Agree
5 = Strongly Agree

Evaluation Criteria	**1**	**2**	**3**	**4**	**5**
The application is easy to use and navigate.					
The image capture process using the device camera is straightforward.					
The disease diagnosis results are delivered quickly.					
The treatment recommendations are easy to understand.					
I would recommend this system to other farmers.					

SECTION C: Open-Ended Feedback

7. Did you encounter any defects, crashes, or issues during testing? If yes, please describe them.
8. What features of the application did you find most useful?
9. What improvements would you suggest to make the system more suitable for daily farming activities?